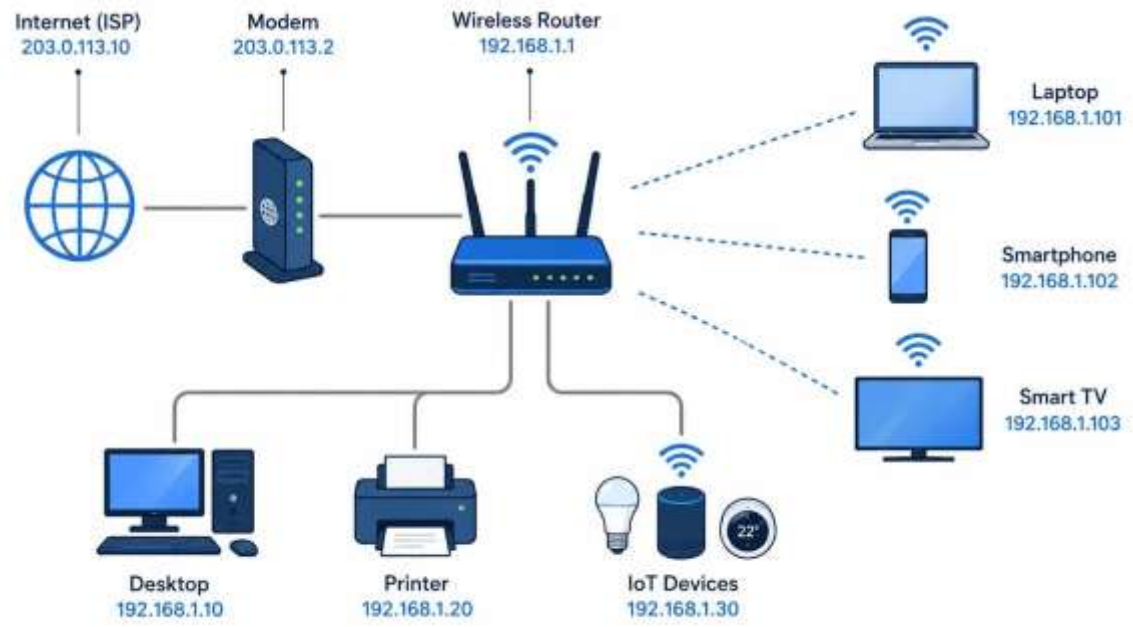
IP Routing And Routing Protocols

Tasks

1.Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.

ANS:



1. Mobile phone connects to the Wi-Fi router.
2. Laptop connects to the same router.
3. Smart TV also connects to the router.
4. The router connects all devices and provides internet access.
5. Every device has a different IP address.

2.Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.

Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.

ANS:

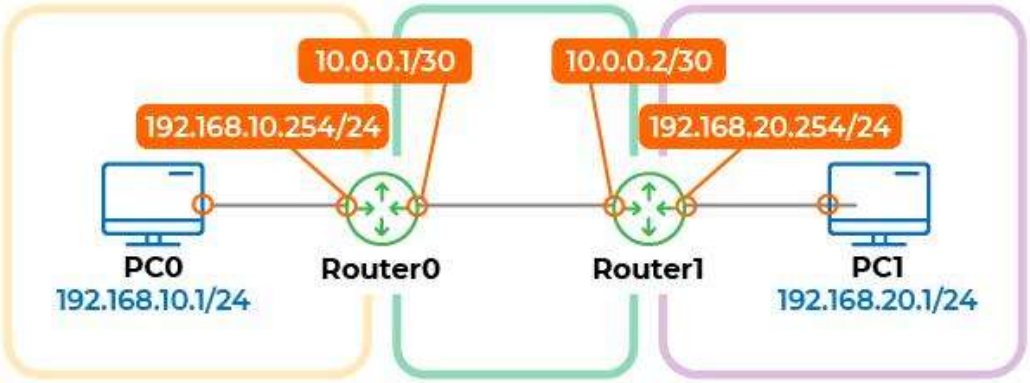
Manual IP Routing

Aim: To understand how data travels from your laptop to your friend's computer in a different network.

Step 1: Example Network
Suppose:

- Your laptop IP: 192.168.1.10
- Your router IP: 192.168.1.1
- Friend's computer IP: 192.168.2.20
- Friend's router IP: 192.168.2.1

Step 2: Network Diagram



Step 3: Routing Table

Device	Destination Network	Next Hop
Your Laptop	192.168.2.0/24	192.168.1.1
Router 1	192.168.2.0/24	Router 2
Router 2	192.168.2.0/24	Direct Connection

Step 4: How Data Travels

1. You send a message from your laptop to your friend's computer.
2. Your laptop checks the friend's IP address.
3. It finds that your friend is in a different network.
4. Your laptop sends the packet to your router (192.168.1.1).
5. Router 1 checks its routing table and forwards the packet to Router 2.
6. Router 2 sends the packet to your friend's computer (192.168.2.20).
7. Your friend's computer receives the packet.

Step 5: Routing Decision
The router checks its routing table to find the correct path for the packet.
It uses the destination IP address to decide where to send the data.

Real-Life Use
When you send a file or message to someone on another network, routers help deliver the data to the correct computer.

3.Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.

ANS:

Routing Table for Food Delivery Network (Like Zomato)

Aim: To create a routing table for three food delivery branches in different cities.

Step 1: Network Details

Suppose a food delivery company has three branches:

Branch	City	Network Address
Branch 1	Ahmedabad	192.168.1.0
Branch 2	Rajkot	192.168.2.0
Branch 3	Surat	192.168.3.0

Step 2: Routing Table

City	Network Address	Subnet Mask	Next Hop	Interface
Ahmedabad	192.168.1.0	255.255.255.0	Direct	LAN 1
Rajkot	192.168.2.0	255.255.255.0	10.0.0.2	WAN 1
Surat	192.168.3.0	255.255.255.0	10.0.0.3	WAN 2

Step 3: Explanation (Easy Words)

1. Ahmedabad branch has its own network.
2. Rajkot branch has a different network.
3. Surat branch has another network.
4. The router checks the routing table.
5. It sends data to the correct branch using the next hop and interface.

Real-Life Use

- Customers order food from different cities.
- Restaurants receive order details.
- Delivery partners receive delivery information.
- Routers help send data between all three branches.

4.Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).

ANS:

Static Routing vs Dynamic Routing

Aim: To understand the difference between static routing and dynamic routing.

1. Static Routing

Static routing means the network admin manually sets the path for data.

Example: A small restaurant office uses a fixed network path.

Advantages (Pros):

1. Easy to set up in a small network.
2. More control over the network.
3. Does not use much network power.

Disadvantages (Cons):

1. Admin must change routes manually.
2. Not suitable for large networks.
3. If a route stops working, it may not find another path automatically.

2. Dynamic Routing

Dynamic routing automatically finds the best path for data using protocols like RIP and OSPF.

Example: A Zomato-style app uses different network paths to connect branches in different cities.

Advantages (Pros):

1. Automatically finds routes.
2. Can find another path if one path fails.
3. Useful for large networks.

Disadvantages (Cons):

1. More difficult to set up.
2. Uses more network resources.
3. Needs proper configuration and maintenance.

3. Comparison Table

Static Routing	Dynamic Routing
Routes are set manually.	Routes are selected automatically.

Best for small networks.	Best for large networks.
Less network resource usage.	More network resource usage.
Does not automatically find another route.	Can find another route.
Example: Small restaurant office.	Example: Large food delivery company.

Real-Life Use

A food delivery company like Zomato has branches in many cities. Dynamic routing helps connect all branches. If one network path fails, it can find another path automatically.

5.Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.

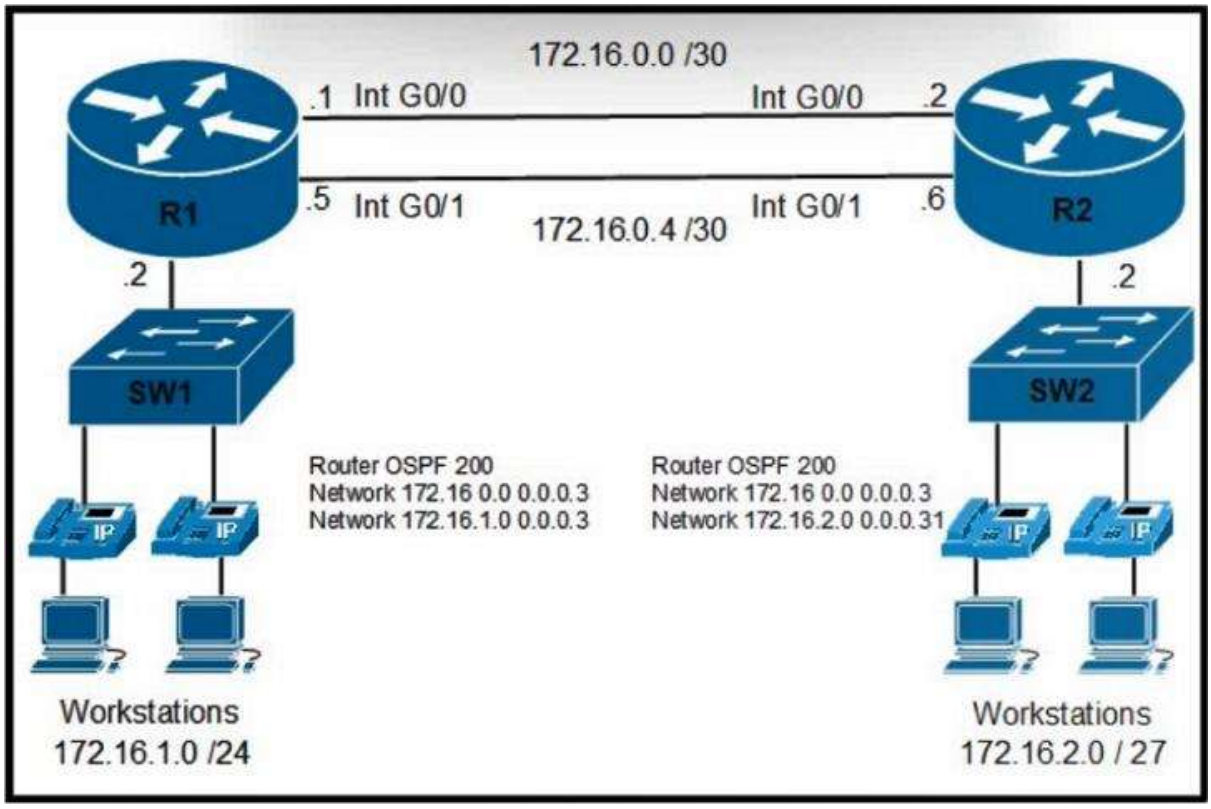
Hint: Paste the AI-generated config and annotate each part with a short description.

ANS:

OSPF Routing Configuration Between Two Routers

Aim: To configure OSPF routing between two routers and understand each command.

Step 1: Network Diagram



Suppose we have two routers:

- Router 1 IP: 192.168.1.1
- Router 2 IP: 192.168.2.1
- Connection between routers: 10.0.0.1 and 10.0.0.2

Step 2: Router 1 Configuration

enable

configure terminal

router ospf 1

network 192.168.1.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

Explanation:

Command	Meaning
enable	Starts administrator mode.
configure terminal	Opens configuration mode.
router ospf 1	Starts OSPF routing.
network 192.168.1.0	Adds the first network to OSPF.
network 10.0.0.0	Adds the router connection network.
area 0	Places the network in OSPF area 0.

Step 3: Router 2 Configuration

enable

configure terminal

router ospf 1

network 192.168.2.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

Explanation:

- enable – Starts administrator mode.
- configure terminal – Opens configuration mode.
- router ospf 1 – Starts OSPF.

- network 192.168.2.0 – Adds Router 2's network.
- network 10.0.0.0 – Adds the connection between routers.
- area 0 – Uses OSPF area 0.

Step 4: Check OSPF Connection

Use these commands:

show ip route

show ip ospf neighbor

- show ip route – Shows the routing table.
- show ip ospf neighbor – Checks whether both routers are connected through OSPF.

Real-Life Use

- Large companies use OSPF to connect different offices.
- Colleges use routing to connect different computer labs.
- Food delivery companies can use routing to connect branches in different cities.